LEARNING HUB MASTER REFERENCE SERIES • VOLUME 03

Linear Algebra: Vector Spaces, Subspaces & The Four Fundamental Subspaces

A rigorous, publication-grade manual synthesizing the 8 vector space axioms, nullspace elimination, column space reachability, complete affine solutions $x = x_p + x_n$, reduced row echelon block factorization, and Gilbert Strang's celebrated Big Picture of Linear Algebra.

Core Theoretical Pillars Covered:

- **Vector Spaces & Subspaces:** The 8 axioms, subspace closure criteria, matrix spaces $M_{m \times n}$, and function spaces.
- **The Nullspace N(A):** Solving $Ax = 0$, pivot variables vs. free variables, and the special solutions formula $[-F; I]$.
- **The Column Space C(A):** Solvability conditions, particular solutions x_p, and the complete solution hyperplane.
- **The 4 Possibilities for Ax = b:** Complete taxonomy based on rank r, row dimension m, and column dimension n.
- **Fundamental Theorem of Linear Algebra (Part 1):** The 4 fundamental subspaces, dimensions, and orthogonal complements.

Curriculum & Benchmark Authority: Gilbert Strang (MIT 18.06 & 5th Ed.), Kenji Hiranabe (The Art of Linear Algebra)
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Part I: Vector Spaces & Subspaces

1.1 The Definition & 8 Fundamental Axioms of a Vector Space

The concept of a **vector space** represents the grand conceptual leap from the concrete geometry of arrows in two and three dimensions to an abstract, axiomatic algebraic framework. A vector space V over the real numbers $\mathbb{R}$ is a collection of objects (termed *vectors*) equipped with two fundamental operations:

1. **Vector Addition (+):** An operation combining any two vectors $\mathbf{u}, \mathbf{v} \in V$ to produce a unique sum $\mathbf{u} + \mathbf{v} \in V$.
2. **Scalar Multiplication (·):** An operation scaling any vector $\mathbf{u} \in V$ by a real scalar $c \in \mathbb{R}$ to yield $c\mathbf{u} \in V$.

For V to qualify as a vector space, these two operations must satisfy exactly eight algebraic axioms for all vectors $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and all scalars $c, d \in \mathbb{R}$:

▶ The 8 Formal Axioms of a Vector Space

Category	Axiom Name & Formula	Physical & Computational Meaning
Axioms for Addition	**1. Commutativity:** $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$	The order of vector addition does not alter the resultant vector (parallelogram rule).
	2. Associativity: $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$	Grouping in multi-vector sums is irrelevant; parentheses can be omitted safely.
	3. Zero Vector Existence: $\exists\, \mathbf{0} \in V$ such that $\mathbf{u} + \mathbf{0} = \mathbf{u}$	There exists a unique additive identity element (the origin) that leaves any vector unchanged.
	4. Additive Inverse: $\forall\, \mathbf{u} \in V, \exists\, (-\mathbf{u})$ with $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$	Every vector can be completely cancelled by its exact opposite direction.
Axioms for Scaling	**5. Vector Distributivity:** $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$	Scalar scaling distributes over vector addition (uniform geometric scaling).
	6. Scalar Distributivity: $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$	Scaling distributes over ordinary scalar arithmetic.
	7. Scalar Associativity: $c(d\mathbf{u}) = (cd)\mathbf{u}$	Successive scaling by scalars d then c equals single scaling by the product cd.
	8. Unitary Scalar Identity: $1 \cdot \mathbf{u} = \mathbf{u}$	Scaling by the real number 1 acts as the multiplicative identity on vectors.

Notice that closure under addition ($\mathbf{u} + \mathbf{v} \in V$) and closure under scalar multiplication ($c\mathbf{u} \in V$) ensure that operations never produce results outside the universe V.

1.2 Subspaces: The Closure Criteria & The Origin Requirement

A **subspace** S of a vector space V is a non-empty subset that is itself a vector space under the same operations already defined on V. Because the eight axioms already hold throughout V, they automatically

inherit to any subset. Therefore, verifying whether S is a subspace requires checking only two closure conditions:

▶ The Subspace Test (The Two Golden Rules)

A subset $S \subseteq V$ is a subspace if and only if it satisfies:

1. **Closure under Addition:** For all $\mathbf{u}, \mathbf{v} \in S$, the sum $\mathbf{u} + \mathbf{v} \in S$.
2. **Closure under Scalar Multiplication:** For all $\mathbf{u} \in S$ and all scalars $c \in \mathbb{R}$, $c\mathbf{u} \in S$.

Equivalent Single-Step Linear Combination Test:

$$\forall\, \mathbf{u}, \mathbf{v} \in S \quad \text{and} \quad \forall\, c, d \in \mathbb{R} \quad \Rightarrow \quad c\mathbf{u} + d\mathbf{v} \in S$$

⚠ The Origin Requirement: Why Zero MUST Belong to Every Subspace

Setting the scalar $c = 0$ in Rule 2 immediately implies:

$$0 \cdot \mathbf{u} = \mathbf{0} \in S$$

Every legitimate subspace must contain the zero vector 0! If a set misses the origin, it is automatically disqualified from being a subspace without any further calculation. Any line or plane in $\mathbb{R}^3$ that does not pass through (0, 0, 0) is NEVER a subspace.

1.3 Subspaces in $\mathbb{R}^n$ and Classic Counterexamples

In Euclidean space $\mathbb{R}^3$, the only subsets that qualify as true subspaces are:

- **The Zero Subspace {0}:** The single point (0, 0, 0), dimension 0.
- **Lines through the Origin:** Sets of the form $S = \{c\mathbf{v} \mid c \in \mathbb{R}\}$ with $\mathbf{v} \neq \mathbf{0}$, dimension 1.
- **Planes through the Origin:** Sets defined by $a\,x + b\,y + c\,z = 0$, dimension 2.
- **The Entire Space $\mathbb{R}^3$:** The full space itself, dimension 3.

◆ Classic Counterexamples: Traps to Avoid

1. Shifted Planes (Affine Planes): The plane $x + 2y - z = 4$ does NOT contain (0, 0, 0) because $0 + 0 - 0 \neq 4$. Thus, it is NOT a subspace.
2. The First Quadrant in $\mathbb{R}^2$: The set $Q_1 = \{[x, y]^T \mid x \geq 0, y \geq 0\}$ contains $\mathbf{0}$ and is closed under addition. However, it fails scalar multiplication: if $\mathbf{u} = [1, 2]^T \in Q_1$, then $(-1)\mathbf{u} = [-1, -2]^T \notin Q_1$.
3. The Union of Two Subspaces ($S_1 \cup S_2$): Consider the x-axis and the y-axis in $\mathbb{R}^2$. Both are 1D subspaces containing $\mathbf{0}$. But their union is NOT a subspace! Take $\mathbf{u} = [1, 0]^T$ on the x-axis and $\mathbf{v} = [0, 1]^T$ on the y-axis. Their sum $\mathbf{u} + \mathbf{v} = [1, 1]^T$ lies in neither axis! (Theorem: $S_1 \cup S_2$ is a subspace iff one subspace contains the other).
4. The Intersection of Two Subspaces ($S_1 \cap S_2$): In sharp contrast to unions, the intersection of any two subspaces is **always** a subspace!

1.4 Abstract Vector Spaces: Matrix Spaces & Function Spaces

The true power of linear algebra is that vectors do not have to be numerical column arrays. Matrices, polynomials, and continuous functions obey the exact same 8 axioms:

- **Matrix Spaces $\mathbf{M_{m \times n}}$:** The set of all $m \times n$ real matrices forms a vector space of dimension $m \times n$. Within square matrices $M_{n \times n}$:

- *Symmetric Matrices ($A^T = A$):* Subspace of dimension $n(n + 1)/2$.
- *Skew-Symmetric Matrices ($A^T = -A$):* Subspace of dimension $n(n - 1)/2$.
- *Upper Triangular Matrices:* Subspace of dimension $n(n + 1)/2$.
- *Diagonal Matrices:* Subspace of dimension n.

- **Polynomial Spaces P_n:** The set of all real polynomials of degree $\le n$: $p(x) = a_0 + a_1 x + \dots + a_n x^n$ forms a vector space of dimension $n + 1$ with standard basis $\{1, x, x^2, \dots, x^n\}$.
- **Solutions to Homogeneous Linear Differential Equations:** The set of all smooth functions satisfying $y'' + \omega^2 y = 0$ forms a 2-dimensional vector space spanned by $\{\cos(\omega t), \sin(\omega t)\}$. Every solution is a linear combination $y(t) = c_1 \cos(\omega t) + c_2 \sin(\omega t)$.

Figure 3.4: Subspaces vs. Non-Subspaces in $\mathbb{R}^3$: The Origin & Linearity Test

A subset $S \subseteq \mathbb{R}^3$ is a subspace if and only if it contains 0 and is closed under both addition and scalar scaling

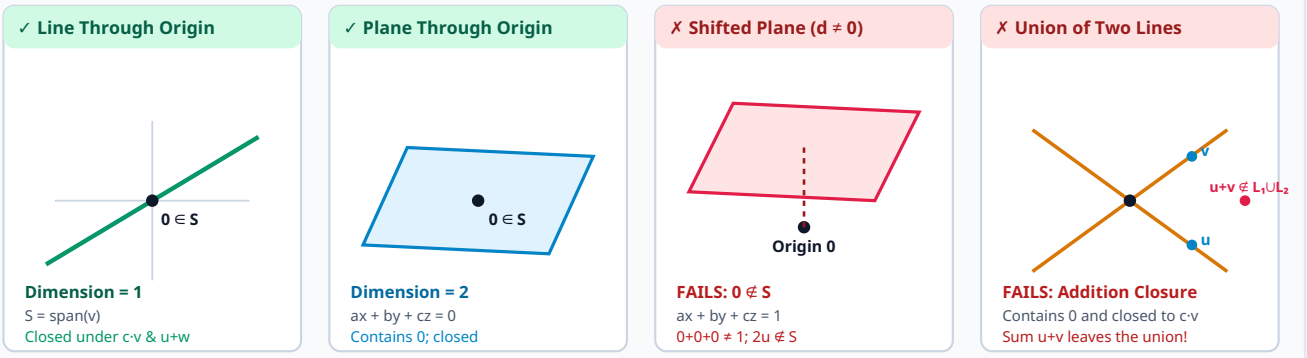


Figure 3.4: Visualizing Subspaces vs. Non-Subspaces in $\mathbb{R}^3$. Lines and planes through the origin satisfy both addition and scaling closures. Shifted planes miss 0, while unions fail vector addition closure.

Part II: The Nullspace N(A) & Solving Ax = 0

2.1 The Nullspace Definition & Subspace Proof

For any $m \times n$ matrix A, the **nullspace** $N(A)$ (also known as the *kernel* of transformation $\mathbf{x} \mapsto A\mathbf{x}$) consists of all input vectors $\mathbf{x} \in \mathbb{R}^n$ that are mapped to the zero vector $\mathbf{0} \in \mathbb{R}^m$:

▶ Definition of the Nullspace N(A)

$$N(A) = \{ \mathbf{x} \in \mathbb{R}^n \mid A\mathbf{x} = \mathbf{0} \} \subseteq \mathbb{R}^n$$

Rigorous Proof that N(A) is a Subspace of $\mathbb{R}^n$:

1. **Contains 0:** Since $A\mathbf{0} = \mathbf{0}$, the zero vector $\mathbf{0} \in N(A)$.
2. **Closed under Addition:** Suppose $\mathbf{u}, \mathbf{v} \in N(A)$. Then $A\mathbf{u} = \mathbf{0}$ and $A\mathbf{v} = \mathbf{0}$. By linearity:

$$A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v} = \mathbf{0} + \mathbf{0} = \mathbf{0} \quad \Rightarrow \quad \mathbf{u} + \mathbf{v} \in N(A)$$

3. **Closed under Scaling:** For any scalar $c \in \mathbb{R}$:

$$A(c\mathbf{u}) = c(A\mathbf{u}) = c(\mathbf{0}) = \mathbf{0} \quad \Rightarrow \quad c\mathbf{u} \in N(A)$$

Therefore, $N(A)$ is unconditionally a subspace of the domain $\mathbb{R}^n$.

2.2 Pivot Columns vs. Free Variables in Echelon Form

To find $N(A)$ explicitly, we perform Gaussian elimination on A to reach an upper triangular or echelon form U, and then continue upward elimination to reach the **Reduced Row Echelon Form** $R = \text{rref}(A)$. Row operations do *not* alter the solutions to $A\mathbf{x} = \mathbf{0}$:

$$A\mathbf{x} = \mathbf{0} \quad \Leftrightarrow \quad U\mathbf{x} = \mathbf{0} \quad \Leftrightarrow \quad R\mathbf{x} = \mathbf{0}$$

When row reduction terminates, every column of R falls into one of two mutually exclusive categories:

- **Pivot Columns (Rank r):** Columns that contain a leading entry (a pivot 1). The corresponding variables are **pivot variables**. There are exactly r pivot variables.
- **Free Columns (Nullity n − r):** Columns that do *not* contain pivots. The corresponding variables are **free variables**. There are exactly $n - r$ free variables.

2.3 Special Solutions to Ax = 0

Free variables can be assigned any arbitrary real values. Each free variable gives rise to an independent **special solution** $\mathbf{s}_j$. The algorithmic recipe to compute all special solutions is:

1. Set one free variable equal to 1, and set all other free variables equal to 0.
2. Solve $R\mathbf{x} = \mathbf{0}$ for the r pivot variables by direct back-substitution.
3. Repeat this process for every free variable.

This generates exactly $n - r$ special solutions $\{\mathbf{s}_1, \mathbf{s}_2, \ldots, \mathbf{s}_{n-r}\}$. Every vector in $N(A)$ is a unique linear combination of these special solutions:

$$\mathbf{x}_n = c_1\,\mathbf{s}_1 + c_2\,\mathbf{s}_2 + \ldots + c_{n-r}\,\mathbf{s}_{n-r}, \qquad \forall\, c_j \in \mathbb{R}$$

2.4 Reduced Row Echelon Form R and the Nullspace Matrix N = [-F; I]

Suppose that the r pivot columns of A appear first (or are grouped together via column permutation). Then the Reduced Row Echelon Form R has the canonical block partition:

$$R = \begin{bmatrix} I_{r\times r} & F_{r\times(n-r)} \\ 0_{(m-r)\times r} & 0_{(m-r)\times(n-r)} \end{bmatrix}$$

where I is the $r \times r$ identity matrix, and F contains the free coefficients. We assemble all $n - r$ special solutions as the columns of the **Nullspace Matrix** N:

$$N = \begin{bmatrix} -F_{r\times(n-r)} \\ I_{(n-r)\times(n-r)} \end{bmatrix}$$

▶ The Nullspace Matrix Cancellation Proof: RN = 0

Multiplying block matrix R by block matrix N:

$$R\,N = \begin{vmatrix} I & F \\ 0 & 0 \end{vmatrix} \begin{vmatrix} -F \\ I \end{vmatrix} = \begin{vmatrix} I(-F) + F(I) \\ 0(-F) + 0(I) \end{vmatrix} = \begin{vmatrix} -F + F \\ 0 \end{vmatrix} = \begin{vmatrix} 0 \\ 0 \end{vmatrix} = \mathbf{0}!$$

Since $R\,N = \mathbf{0}$, every column of N satisfies $R\mathbf{s}_j = \mathbf{0}$, and hence $A\mathbf{s}_j = \mathbf{0}$. Furthermore, because the bottom block of N is the identity matrix $I_{(n-r)}$, the columns of N are strictly linearly independent, proving that they form an exact **basis** for $N(A)$.

Figure 3.3: Matrix Block Structure: RREF Matrix R and Nullspace Matrix N

When pivot columns come first, RN = 0 reveals all n − r special solutions directly in closed form

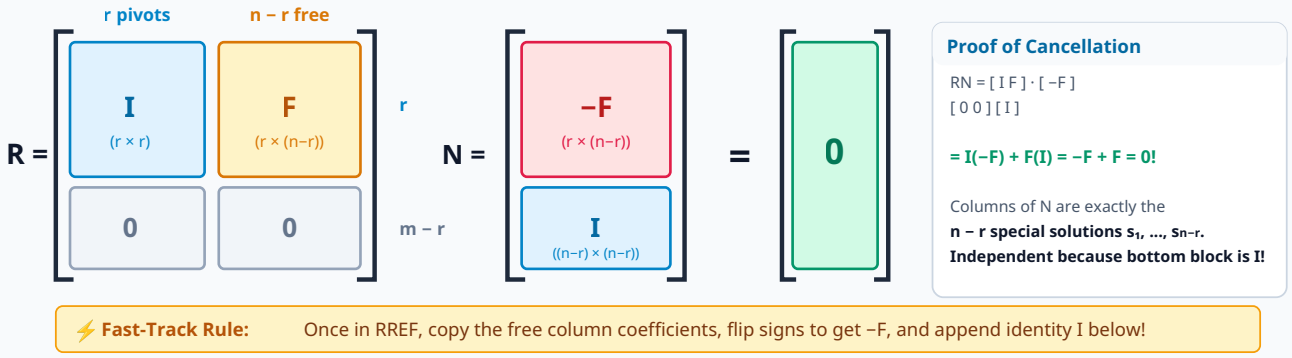


Figure 3.3: Canonical block structure of R and the Nullspace Matrix N. The product RN = 0 proves that the columns of N are precisely the special solutions to Ax = 0.

⚡ FAST-TRACK 1 Reading Free Variables & Pivots from R in 5 Seconds

Exam Shortcut: Once a matrix is in RREF, do NOT set up long algebraic equations!

1. Count the staircase leading 1s: their column numbers = **Pivot Variables**.
2. The remaining column indices = **Free Variables**.
3. Nullity = Total columns n − Number of pivots r.
4. If there are k free variables, you will have exactly k independent basis vectors in $N(A)$!

⚡ FAST-TRACK 2 The Instant [-F; I] Nullspace Formula

Never back-substitute manually again! For any RREF where pivots are in columns 1 through r:

$$R = \begin{vmatrix} 1 & 0 & f_{11} & f_{12} \\ 0 & 1 & f_{21} & f_{22} \end{vmatrix} \quad \Rightarrow \quad N = \begin{vmatrix} -f_{11} & -f_{12} \\ -f_{21} & -f_{22} \\ 1 & 0 \\ 0 & 1 \end{vmatrix}$$

Simply transcribe the free entries, flip their signs ($-F$), and place an identity matrix (I) on the rows corresponding to the free variables!

Part III: The Column Space C(A) & The Complete Solution

3.1 The Column Space C(A) & Solvability Condition b ∈ C(A)

The **column space** *C(A)* (also known as the *range* or *image* of *A*) is the subspace of $\mathbb{R}^m$ spanned by all *n* column vectors of *A*:

$$C(A) = \text{span}\{ \mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n \} = \{ \mathbf{b} \in \mathbb{R}^m \mid \mathbf{b} = A\mathbf{x} \text{ for some } \mathbf{x} \in \mathbb{R}^n \} \subseteq \mathbb{R}^m$$

Since $A\mathbf{x}$ is precisely a linear combination of the columns of *A*: $A\mathbf{x} = x_1 \mathbf{a}_1 + x_2 \mathbf{a}_2 + \ldots + x_n \mathbf{a}_n$, we obtain the foundational existence theorem:

▶ Solvability Condition for Ax = b

The linear system $A\mathbf{x} = \mathbf{b}$ has a solution if and only if the right-hand side vector **b** lies in the column space of *A*:

$$A\mathbf{x} = \mathbf{b} \text{ is solvable} \quad \Leftrightarrow \quad \mathbf{b} \in C(A)$$

Row Reduction Test: When reducing the augmented matrix $[A \mid \mathbf{b}] \to [R \mid \mathbf{d}]$, every row of zeros in *R* MUST have a corresponding zero in **d**. If any row reads $[\, 0\ 0 \ldots 0 \mid d_k \,]$ with $d_k \neq 0$, the system is inconsistent ($0 \neq d_k$).

3.2 The Particular Solution x_p

When $A\mathbf{x} = \mathbf{b}$ is solvable, we isolate a single specific solution $\mathbf{x}_p$ known as the **particular solution**. The most systematic and fail-safe method to find $\mathbf{x}_p$ is:

Recipe for Particular Solution $\mathbf{x}_p$:

Set **all free variables strictly equal to 0** in $R\mathbf{x} = \mathbf{d}$.

The remaining *r* pivot variables then equal the corresponding entries of **d** directly!

3.3 The Complete Solution: x = x_p + x_n

The complete solution to $A\mathbf{x} = \mathbf{b}$ combines the particular solution with the entire nullspace:

▶ The Complete Solution Theorem

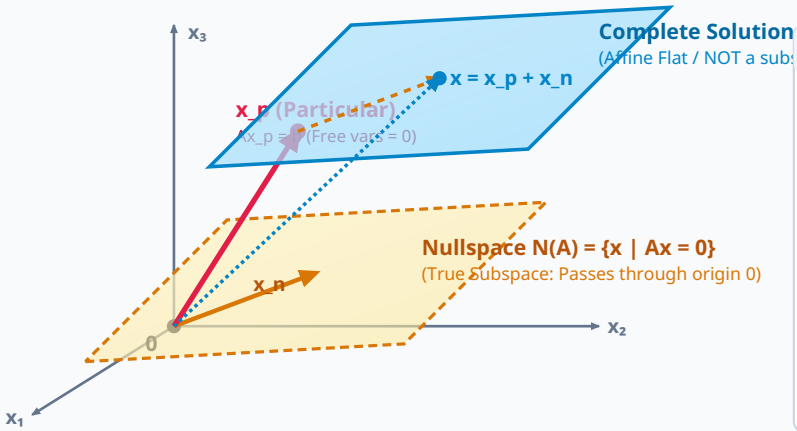
Every solution to the inhomogeneous system $A\mathbf{x} = \mathbf{b}$ can be written uniquely as:

$$\mathbf{x}_{\text{complete}} = \mathbf{x}_p + \mathbf{x}_n = \mathbf{x}_p + c_1 \mathbf{s}_1 + c_2 \mathbf{s}_2 + \ldots + c_{n-r} \mathbf{s}_{n-r}$$

where $\mathbf{x}_p$ is any fixed particular solution ($A\mathbf{x}_p = \mathbf{b}$), and $\mathbf{x}_n \in N(A)$ is the general solution to $A\mathbf{x}_n = \mathbf{0}$.

Figure 3.2: Geometric Anatomy of the Complete Solution x = x_p + x_n

The solution set to Ax = b is an affine plane parallel to the nullspace N(A), shifted away from origin by x_p



Key Geometric Properties

- **Homogeneous Ax = 0:** Passes through origin (Subspace).
- **Inhomogeneous Ax = b:** Shifted plane parallel to N(A).
- **Not a Subspace:** Does not contain 0; c·x fails.
- **Vector Differences:** If x, x' solve Ax = b, then: **x − x' ∈ N(A)** since A(x − x') = b − b = 0!
- **Free Variables:** Dim(N(A)) = n − r directions **along the shifted plane.**

Figure 3.2: Complete Solution Geometry. The solution set to Ax = b forms an affine hyperplane parallel to N(A), shifted away from the origin by the particular solution vector x_p.

Part IV: Independence, Basis, Dimension & The 4 Possibilities

4.1 Linear Independence, Spanning Sets, and Basis Definitions

Three interconnected concepts form the bedrock of dimensional analysis in linear algebra:

▶ Formal Definitions: Independence, Span, and Basis

1. **Linear Independence:** A set of vectors $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ is linearly independent if:

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \ldots + c_k \mathbf{v}_k = \mathbf{0} \quad \Rightarrow \quad c_1 = c_2 = \ldots = c_k = 0$$

   Matrix Test: The columns of matrix $V = [\mathbf{v}_1 \ldots \mathbf{v}_k]$ are independent iff $N(V) = \{\mathbf{0}\}$ (rank $r = k$, no free variables).

2. **Spanning Set:** Vectors $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ span space V if every vector $\mathbf{u} \in V$ can be written as a linear combination: $\mathbf{u} = \sum c_i \mathbf{v}_i$. **Matrix Test:** The columns of V span $\mathbb{R}^m$ iff rank $r = m$ (pivots in every row).

3. **Basis:** A set of vectors $B = \{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ is a **basis** for V if:
   (a) It is linearly independent, **AND** (b) It spans V.
   Equivalently: A basis is a **minimal spanning set** and a **maximal independent set**.

4.2 Dimension & The Rank-Nullity Theorem: r + (n - r) = n

The **dimension** of a vector space V, denoted dim(V), is the unique number of vectors in any basis for V. If V has a basis of k vectors, every other basis for V also contains exactly k vectors.

▶ The Rank-Nullity Theorem (Conservation of Domain Dimension)

For any $m \times n$ matrix A with rank r:

$$\text{rank}(A) + \text{nullity}(A) = n \quad \Leftrightarrow \quad \dim(C(A)) + \dim(N(A)) = n$$

Intuition: The domain $\mathbb{R}^n$ has n degrees of freedom. During matrix multiplication $\mathbf{x} \mapsto A\mathbf{x}$, exactly r dimensions survive in the output column space $C(A)$, while the remaining $n - r$ dimensions are completely flattened to zero in the nullspace $N(A)$.

4.3 The 4 Possibilities for Ax = b (Square, Wide, Tall, Deficient)

The rank r of an $m \times n$ matrix cannot exceed the row count m or column count n: $r \le \min(m, n)$. This gives rise to exactly **four fundamental possibilities** for the solvability of $A\mathbf{x} = \mathbf{b}$:

Rank Condition	Matrix Shape	RREF Block Form R	Nullspace N(A)	Column Space C(A)	Solutions to Ax = b
r = m = n Full Rank	Square ($n \times n$) Invertible	$R = I$	dim = 0 $\{\mathbf{0}\}$ only	dim = n Full space $\mathbb{R}^n$	**Exactly 1 unique solution** for every **b**: $\mathbf{x} = A^{-1}\mathbf{b}$
r = m < n Full Row Rank	Short & Wide (Underdetermined)	$R = [\, I \quad F \,]$ No zero rows	dim = $n - m$ > 0 Free variables	dim = m Full space $\mathbb{R}^m$	**Infinitely many solutions** for every $\mathbf{b} \in \mathbb{R}^m$: $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_n$
r = n < m Full Column Rank	Tall & Thin (Overdetermined)	$R = \begin{bmatrix} I \\ 0 \end{bmatrix}$ Zero rows at bottom	dim = 0 $\{\mathbf{0}\}$ only	dim = $n < m$ Proper subspace	**0 or 1 unique solution:** 1 if $\mathbf{b} \in C(A)$; 0 if $\mathbf{b} \notin C(A)$
r < m and r < n Rank Deficient	General Rectangular (Not full rank)	$R = \begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix}$	dim = $n - r$ > 0 Free variables	dim = $r < m$ Proper subspace	**0 or ∞ solutions:** ∞ if $\mathbf{b} \in C(A)$; 0 if $\mathbf{b} \notin C(A)$

⚡ FAST-TRACK 3 **The 4-Way Rank Solvability Diagnostic Chart**

Instant Exam Decision Rule:

- Full Column Rank ($r = n$) ⇒ Independent columns ⇒ At most **1** solution (0 or 1).
- Full Row Rank ($r = m$) ⇒ Spanning columns ⇒ At least **1** solution (1 or ∞).
- Full Rank ($r = m = n$) ⇒ Independent + Spanning ⇒ Exactly **1** unique solution.
- Deficient Rank ($r < m$, $r < n$) ⇒ Zero rows & free variables ⇒ 0 or ∞ solutions!

⚡ FAST-TRACK 4 **Basis by Inspection & Linear Dependence Traps**

Scan vectors visually for immediate dependence before doing row reduction:

1. **The Pigeonhole Rule:** Any set of k vectors in $\mathbb{R}^n$ with $k > n$ is **guaranteed linearly dependent** (more unknowns than equations ⇒ free variables!).
2. **The Zero Vector Rule:** Any set containing $\mathbf{0}$ is automatically dependent ($0 \cdot \mathbf{0} + 0 \cdot \ldots$ with non-zero weight $1 \cdot \mathbf{0} = \mathbf{0}$).
3. **Proportionality:** If vector $\mathbf{v}_2 = c\,\mathbf{v}_1$, they are instantly dependent.
4. **Original Columns for C(A) Basis:** Never take the pivot columns of R! Row reduction alters the column space. Always extract the corresponding pivot columns of the **original** matrix A!

Part V: The Four Fundamental Subspaces (Strang's Big Picture)

5.1 The Four Fundamental Subspaces Defined

The Fundamental Theorem of Linear Algebra connects four fundamental subspaces associated with every $m \times n$ matrix A:

Subspace Name	Symbol	Ambient Space	Dimension	How to Find an Exact Basis
1. Row Space	$C(A^T)$	$\mathbb{R}^n$	r (Rank)	The r non-zero rows of Reduced Row Echelon Form R (or U).
2. Nullspace	$N(A)$	$\mathbb{R}^n$	$n - r$ (Nullity)	The $n - r$ special solutions obtained from $N = [-F; I]$.
3. Column Space	$C(A)$	$\mathbb{R}^m$	r (Rank)	The r pivot columns of the **original matrix A** (not R!).
4. Left Nullspace	$N(A^T)$	$\mathbb{R}^m$	$m - r$ (Cokernel)	The $m - r$ rows of elimination matrix E that produce zero rows in $EA = R$.

5.2 Orthogonal Complements: $\mathbb{R}^n = C(A^T) \oplus N(A)$ and $\mathbb{R}^m = C(A) \oplus N(A^T)$

Not only do these subspaces live in the domain $\mathbb{R}^n$ and codomain $\mathbb{R}^m$, but they are mutually **orthogonal complements**:

▶ Fundamental Theorem of Linear Algebra (Part 1: Orthogonality)

1. **In Domain $\mathbb{R}^n$:** The nullspace is the orthogonal complement of the row space:

$$N(A) = (C(A^T))^{\perp} \quad \Leftrightarrow \quad \mathbb{R}^n = C(A^T) \oplus N(A)$$

Proof: $A\mathbf{x} = \mathbf{0}$ means that every row of A has dot product zero with $\mathbf{x}$: $(\text{row}_i) \cdot \mathbf{x} = 0$. Thus $\mathbf{x}$ is perpendicular to every row and to all their linear combinations.

2. **In Codomain $\mathbb{R}^m$:** The left nullspace is the orthogonal complement of the column space:

$$N(A^T) = (C(A))^{\perp} \quad \Leftrightarrow \quad \mathbb{R}^m = C(A) \oplus N(A^T)$$

Proof: $A^T\mathbf{y} = \mathbf{0} \Leftrightarrow \mathbf{y}^TA = \mathbf{0}^T$ means that $\mathbf{y}$ has dot product zero with every column of A: $(\text{col}_j) \cdot \mathbf{y} = 0$.

5.3 The Big Picture: True Invertible Mapping between Row & Column Spaces

Gilbert Strang's celebrated "Big Picture" reveals the elegant geometric action of matrix A on any vector $\mathbf{x} \in \mathbb{R}^n$:

1. Every input $\mathbf{x} \in \mathbb{R}^n$ splits uniquely into an orthogonal sum: $\mathbf{x} = \mathbf{x}_r + \mathbf{x}_n$, where $\mathbf{x}_r \in C(A^T)$ and $\mathbf{x}_n \in N(A)$.
2. When A multiplies $\mathbf{x}$, the nullspace component completely vanishes: $A\mathbf{x}_n = \mathbf{0}$.
3. The row space component $\mathbf{x}_r$ is mapped directly and uniquely into the column space: $A\mathbf{x}_r = \mathbf{b} \in C(A)$.
4. **The Invertible Core:** Restricted to the row space $C(A^T)$, A acts as an **invertible, one-to-one, and onto transformation** mapping r-dimensional $C(A^T)$ onto r-dimensional $C(A)$!

Figure 3.1: The Big Picture of Linear Algebra — The Four Fundamental Subspaces

Direct sum decomposition into orthogonal complements: $\mathbb{R}^n = C(A^T) \oplus N(A)$ and $\mathbb{R}^m = C(A) \oplus N(A^T)$

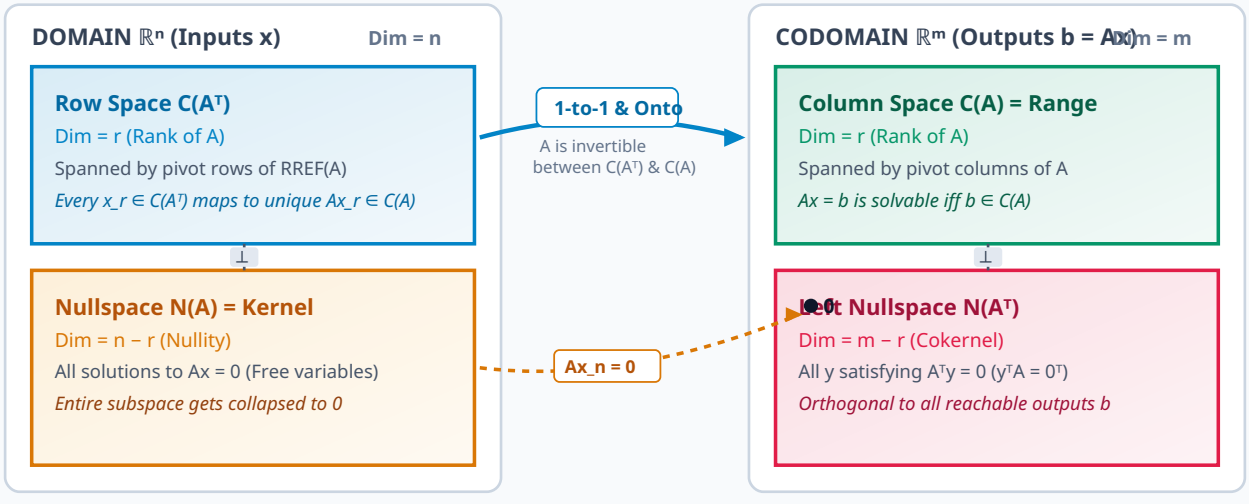


Figure 3.1: Gilbert Strang's Big Picture of Linear Algebra. Domain $\mathbb{R}^n$ splits into Row Space and Nullspace; Codomain $\mathbb{R}^m$ splits into Column Space and Left Nullspace. Matrix A maps $C(A^T)$ invertibly onto $C(A)$ and squashes $N(A)$ to 0.

Part VI: Publication-Grade Worked Problems

Worked Problem 3.1: Complete 4-Subspace Audit for a 3×4 Rectangular Matrix

Problem Statement: Consider the rectangular matrix $A \in \mathbb{R}^{3\times 4}$:

$$A = \begin{vmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 3 & 1 & 1 \end{vmatrix}$$

Perform Gaussian elimination to find $R = \text{rref}(A)$. Determine rank r, identify pivot and free variables, and find an explicit basis and exact dimension for all four fundamental subspaces: $C(A)$, $N(A)$, $C(A^T)$, and $N(A^T)$. Verify the Rank-Nullity Theorem and mutual orthogonality.

Step 1: Row Reduction to Reduced Row Echelon Form R

Apply elementary row operations to A:

$$\begin{vmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 3 & 1 & 1 \end{vmatrix} \xrightarrow{R_3 \to R_3 - R_1} \begin{vmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \end{vmatrix}$$

$$\xrightarrow[R_1 \to R_1 - 2R_2]{R_3 \to R_3 - R_2} R = \begin{vmatrix} \mathbf{1} & 0 & -2 & 1 \\ 0 & \mathbf{1} & 1 & 0 \\ 0 & 0 & 0 & 0 \end{vmatrix}$$

Rank: $r = 2$ (pivots in column 1 and column 2). **Free Variables:** x_3 and x_4 (columns 3 and 4).

Step 2: Basis and Dimension for the Column Space C(A) ⊆ ℝ³

The pivot columns are columns 1 and 2. We extract the corresponding **original** columns of A:

$$\text{Basis for } C(A) = \left\{ \begin{vmatrix} 1 \\ 0 \\ 1 \end{vmatrix}, \begin{vmatrix} 2 \\ 1 \\ 3 \end{vmatrix} \right\}, \quad \dim(C(A)) = r = 2$$

Step 3: Basis and Dimension for the Row Space C(Aᵀ) ⊆ ℝ⁴

The row space is spanned by the $r = 2$ non-zero rows of R:

$$\text{Basis for } C(A^T) = \left\{ \begin{vmatrix} 1 \\ 0 \\ -2 \\ 1 \end{vmatrix}, \begin{vmatrix} 0 \\ 1 \\ 1 \\ 0 \end{vmatrix} \right\}, \quad \dim(C(A^T)) = r = 2$$

Step 4: Basis and Dimension for the Nullspace N(A) ⊆ ℝ⁴

Using the special solutions formula or $R\mathbf{x} = \mathbf{0}$:

- Set $x_3 = 1$, $x_4 = 0 \Rightarrow x_1 - 2(1) + 1(0) = 0 \Rightarrow x_1 = 2$; $x_2 + 1(1) = 0 \Rightarrow x_2 = -1$. $\Rightarrow \mathbf{s}_1 = [\ 2, -1, 1, 0\ ]^T$.
- Set $x_3 = 0$, $x_4 = 1 \Rightarrow x_1 - 2(0) + 1(1) = 0 \Rightarrow x_1 = -1$; $x_2 + 1(0) = 0 \Rightarrow x_2 = 0$. $\Rightarrow \mathbf{s}_2 = [\ -1, 0, 0, 1\ ]^T$.

$$\text{Basis for } N(A) = \left\{ \begin{bmatrix} 2 \\ -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}, \quad \dim(N(A)) = n - r = 4 - 2 = 2$$

Step 5: Basis and Dimension for the Left Nullspace N(Aᵀ) ⊆ ℝ³

We solve $A^T\mathbf{y} = \mathbf{0} \Leftrightarrow \mathbf{y}^T A = \mathbf{0}^T$. Row 3 of A was obtained during elimination as Row 1 + Row 2. Thus: Row 1 + Row 2 − Row 3 = $\mathbf{0}^T$. Hence $\mathbf{y} = [\ 1, 1, -1\ ]^T$:

$$\text{Basis for } N(A^T) = \left\{ \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \right\}, \quad \dim(N(A^T)) = m - r = 3 - 2 = 1$$

Step 6: Mathematical Sanity & Orthogonality Checks

• **Rank-Nullity Domain:** $\dim(C(A^T)) + \dim(N(A)) = 2 + 2 = 4 = n$. ✓
• **Rank-Nullity Codomain:** $\dim(C(A)) + \dim(N(A^T)) = 2 + 1 = 3 = m$. ✓
• **Orthogonality C(Aᵀ) ⊥ N(A):**
Row 1 · $\mathbf{s}_1 = 1(2) + 0(-1) + (-2)(1) + 1(0) = 2 - 2 = 0$. ✓
Row 2 · $\mathbf{s}_1 = 0(2) + 1(-1) + 1(1) + 0(0) = -1 + 1 = 0$. ✓
• **Orthogonality C(A) ⊥ N(Aᵀ):**
Col 1 · $\mathbf{y} = 1(1) + 0(1) + 1(-1) = 0$; Col 2 · $\mathbf{y} = 2(1) + 1(1) + 3(-1) = 0$. ✓

Worked Problem 3.2: Augmented System Elimination & Complete Solution x = x_p + x_n

Problem Statement: Given the matrix A from Problem 3.1 and right-hand side vector $\mathbf{b} = [3, 1, 4]^T$:

$$\begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 3 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 4 \end{bmatrix}$$

Verify that $\mathbf{b} \in C(A)$, compute particular solution $\mathbf{x}_p$ by setting free variables to zero, and construct the complete solution set in parametric vector form.

Step 1: Augmented Matrix Elimination [A | b] → [R | d]

$$[A \mid \mathbf{b}] = \left|\begin{matrix} 1 & 2 & 0 & 1 & 3 \\ 0 & 1 & 1 & 0 & 1 \\ 1 & 3 & 1 & 1 & 4 \end{matrix}\right| \rightarrow \left|\begin{matrix} 1 & 2 & 0 & 1 & 3 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 4-3=1 \end{matrix}\right|$$

$$\rightarrow \left|\begin{matrix} 1 & 0 & -2 & 1 & 3-2(1)=1 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1-1=0 \end{matrix}\right| = [R \mid \mathbf{d}]$$

Consistency Check: Row 3 reads [0 0 0 0 | 0], meaning 0 = 0. The system is consistent and solvable! Notice also that $\mathbf{y} \cdot \mathbf{b} = 1(3) + 1(1) - 1(4) = 0$, verifying that **b** is strictly orthogonal to the left nullspace!

Step 2: Particular Solution $\mathbf{x_p}$

Set free variables $x_3 = 0$ and $x_4 = 0$. The pivot equations read:

$$x_1 - 2(0) + 1(0) = 1 \Rightarrow x_1 = 1, \quad x_2 + 1(0) = 1 \Rightarrow x_2 = 1$$

$$\mathbf{x}_p = \left|\begin{matrix} 1 \\ 1 \\ 0 \\ 0 \end{matrix}\right|$$

Check: $A\mathbf{x}_p = 1[1, 0, 1]^T + 1[2, 1, 3]^T + 0 + 0 = [3, 1, 4]^T = \mathbf{b}$. Exact!

Step 3: Parametric Complete Solution

Combining $\mathbf{x}_p$ with the nullspace special solutions $\mathbf{s}_1$, $\mathbf{s}_2$ found in Problem 3.1:

$$\mathbf{x} = \left|\begin{matrix} 1 \\ 1 \\ 0 \\ 0 \end{matrix}\right| + c_1 \left|\begin{matrix} 2 \\ -1 \\ 1 \\ 0 \end{matrix}\right| + c_2 \left|\begin{matrix} -1 \\ 0 \\ 0 \\ 1 \end{matrix}\right|, \quad \forall\, c_1, c_2 \in \mathbb{R}$$

Worked Problem 3.3: Full Column Rank 4×2 Matrix, Left Inverse & Left Nullspace

Problem Statement: Consider the tall, thin matrix $A \in \mathbb{R}^{4\times 2}$:

$$A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$$

Demonstrate that A has full column rank. Calculate its unique left inverse $A_{\text{left}}^{-1} = (A^T A)^{-1} A^T$, find a basis for its 2-dimensional left nullspace $N(A^T) \subseteq \mathbb{R}^4$, and determine the exact solvability conditions on $\mathbf{b} = [b_1, b_2, b_3, b_4]^T$.

Step 1: Rank and Gram Matrix A^TA

Since column 2 is not a scalar multiple of column 1, the two columns are linearly independent ⇒ rank $r = 2 = n$. Thus $N(A) = \{\mathbf{0}\}$. Compute A^TA:

$$A^TA = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1+4+0+1 & 0+2+0+1 \\ 0+2+0+1 & 0+1+1+1 \end{bmatrix} = \begin{bmatrix} 6 & 3 \\ 3 & 3 \end{bmatrix}$$

$\det(A^TA) = (6)(3) - (3)(3) = 18 - 9 = 9 \neq 0 \Rightarrow A^TA$ is strictly invertible!

$$(A^TA)^{-1} = \frac{1}{9} \begin{bmatrix} 3 & -3 \\ -3 & 6 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$$

Step 2: Explicit Left Inverse A_{left}^{-1}

$$A_{\text{left}}^{-1} = (A^TA)^{-1} A^T = \frac{1}{3} \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & -1 & 0 \\ -1 & 0 & 2 & 1 \end{bmatrix}$$

Check: $A_{\text{left}}^{-1} A = (1/3)\ [\ 3\ 0\ ;\ 0\ 3\ ] = I_2$. ✓

Step 3: Basis for Left Nullspace $\mathbf{N(A^T)} \subseteq \mathbb{R}^4$

We solve $A^T\mathbf{y} = \mathbf{0}$ for $\mathbf{y} \in \mathbb{R}^4$: [$y_1 + 2y_2 + y_4 = 0, \quad y_2 + y_3 + y_4 = 0$]. With two free variables y_3, y_4:

$$\text{Basis for } N(A^T) = \left\{ \begin{bmatrix} 2 \\ -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \end{bmatrix} \right\}, \quad \dim(N(A^T)) = m - r = 4 - 2 = 2$$

Solvability Condition: $A\mathbf{x} = \mathbf{b}$ has a solution iff $\mathbf{y}_1 \cdot \mathbf{b} = 0$ and $\mathbf{y}_2 \cdot \mathbf{b} = 0$:

$$2b_1 - b_2 + b_3 = 0 \quad \text{and} \quad b_1 - b_2 + b_4 = 0$$

Worked Problem 3.4: Symmetric Matrix Subspace S_3 and Constrained Polynomials P_3

Problem Statement:

(a) Prove that the subset of symmetric matrices $S_3 = \{M \in M_{3\times3} \mid M^T = M\}$ is a subspace of $M_{3\times3}$. Construct an explicit basis and find $\dim(S_3)$.

(b) In the polynomial space $P_3 = \{p(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3\}$, find a basis and the dimension for the subspace W consisting of all polynomials satisfying the boundary conditions: $p(0) = 0$ and $p'(1) = 0$.

Step 1: Symmetric Matrix Subspace S_3

1. **Zero:** $0^T = 0 \in S_3$.
2. **Closure:** For any $A, B \in S_3$ and $c, d \in \mathbb{R}$: $(cA + dB)^T = cA^T + dB^T = cA + dB \in S_3$. Thus S_3 is a subspace.

A 3×3 symmetric matrix has the general structure:

$$M = \begin{vmatrix} a & b & c \\ b & d & e \\ c & e & f \end{vmatrix} = a E_{11} + d E_{22} + f E_{33} + b(E_{12} + E_{21}) + c(E_{13} + E_{31}) + e(E_{23} + E_{32})$$

There are 3 independent diagonal parameters and 3 independent off-diagonal parameters. Hence:

$$\dim(S_3) = \frac{n(n+1)}{2} = \frac{3(4)}{2} = \mathbf{6}$$

Step 2: Constrained Polynomial Subspace $W \subseteq P_3$

Evaluate the two constraints on $p(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3$:

- **Constraint 1:** $p(0) = 0 \Rightarrow a_0 = 0$.
- **Constraint 2:** $p'(t) = a_1 + 2a_2 t + 3a_3 t^2 \Rightarrow p'(1) = a_1 + 2a_2 + 3a_3 = 0 \Rightarrow a_1 = -2a_2 - 3a_3$.

Substituting back into $p(t)$:

$$p(t) = (-2a_2 - 3a_3)t + a_2 t^2 + a_3 t^3 = a_2(t^2 - 2t) + a_3(t^3 - 3t)$$

The polynomials $q_1(t) = t^2 - 2t$ and $q_2(t) = t^3 - 3t$ are linearly independent (different degrees) and span W:

$$\text{Basis for } W = \{\, t^2 - 2t,\quad t^3 - 3t \,\}, \qquad \dim(W) = \mathbf{2}$$

Worked Problem 3.5: Rank-1 Outer Product $A = \mathbf{u}\mathbf{v}^T$ & Subspace Geometries

Problem Statement: Let $\mathbf{u} = [1, 2, 3]^T \in \mathbb{R}^3$ and $\mathbf{v} = [2, -1, 1, 4]^T \in \mathbb{R}^4$. Form the rank-1 matrix $A = \mathbf{u}\mathbf{v}^T \in \mathbb{R}^{3\times4}$. Determine its explicit entry values, find closed-form coordinate bases for all four fundamental subspaces, and verify their geometric dimensions.

Step 1: Matrix Outer Product Construction

$$A = \begin{vmatrix} 1 \\ 2 \\ 3 \end{vmatrix} \begin{bmatrix} 2 & -1 & 1 & 4 \end{bmatrix} = \begin{vmatrix} 2 & -1 & 1 & 4 \\ 4 & -2 & 2 & 8 \\ 6 & -3 & 3 & 12 \end{vmatrix}$$

Notice that Row 2 is 2 × Row 1, and Row 3 is 3 × Row 1. Thus **rank(A) = r = 1**.

Step 2: Column Space C(A) and Row Space C(A^T)

Every column is a scalar multiple of **u**, and every row is a scalar multiple of $\mathbf{v}^T$:

$$C(A) = \text{span}\{\ \mathbf{u}\ \} = \text{span}\{\ [1, 2, 3]^T\ \}, \quad \dim(C(A)) = r = 1 \subseteq \mathbb{R}^3$$
$$C(A^T) = \text{span}\{\ \mathbf{v}\ \} = \text{span}\{\ [2, -1, 1, 4]^T\ \}, \quad \dim(C(A^T)) = r = 1 \subseteq \mathbb{R}^4$$

Step 3: Nullspace N(A) and Left Nullspace N(A^T)

Nullspace: $A\mathbf{x} = \mathbf{u}(\mathbf{v}^T\mathbf{x}) = \mathbf{0}$. Since $\mathbf{u} \neq \mathbf{0}$, this requires $\mathbf{v}^T\mathbf{x} = 0$. Thus $N(A) = \mathbf{v}^\perp$, the 3-dimensional hyperplane in $\mathbb{R}^4$ perpendicular to **v**:

$$2x_1 - x_2 + x_3 + 4x_4 = 0 \quad \Rightarrow \quad \dim(N(A)) = n - r = 4 - 1 = \mathbf{3}$$
$$\text{Basis for } N(A) = \left\{ \begin{vmatrix} 1 \\ 2 \\ 0 \\ 0 \end{vmatrix}, \begin{vmatrix} 0 \\ 1 \\ 1 \\ 0 \end{vmatrix}, \begin{vmatrix} 0 \\ 4 \\ 0 \\ 1 \end{vmatrix} \right\}$$

Left Nullspace: $A^T\mathbf{y} = \mathbf{v}(\mathbf{u}^T\mathbf{y}) = \mathbf{0} \Rightarrow \mathbf{u}^T\mathbf{y} = 0$. Thus $N(A^T) = \mathbf{u}^\perp$, the 2-dimensional plane in $\mathbb{R}^3$ perpendicular to **u**:

$$y_1 + 2y_2 + 3y_3 = 0 \quad \Rightarrow \quad \dim(N(A^T)) = m - r = 3 - 1 = \mathbf{2}$$
$$\text{Basis for } N(A^T) = \left\{ \begin{vmatrix} -2 \\ 1 \\ 0 \end{vmatrix}, \begin{vmatrix} -3 \\ 0 \\ 1 \end{vmatrix} \right\}$$

Appendix: Master Summary Table & Solvability Checklist

Appendix A: Master Reference Table for the Four Fundamental Subspaces

Subspace	Ambient Space	Exact Dimension	Orthogonal Complement	Algorithmic Recipe for Basis
Row Space $C(A^T)$	$\mathbb{R}^n$ (Domain)	r	$N(A)$	Row reduce $A \to R$; take the r non-zero rows of R.
Nullspace $N(A)$	$\mathbb{R}^n$ (Domain)	$n - r$	$C(A^T)$	Set each free variable to 1 (others 0) in $R\mathbf{x} = \mathbf{0}$; transcribe columns of $[-F; I]$.
Column Space $C(A)$	$\mathbb{R}^m$ (Codomain)	r	$N(A^T)$	Locate pivot columns in R; take the matching columns from the **original** matrix A.
Left Nullspace $N(A^T)$	$\mathbb{R}^m$ (Codomain)	$m - r$	$C(A)$	Row reduce $[\ A \mid I\ ] \to [\ R \mid E\ ]$; extract the $m - r$ rows of E corresponding to zero rows of R.

Appendix B: Solvability Decision Tree for $A\mathbf{x} = \mathbf{b}$

▶ Step-by-Step Diagnostic Protocol

1. **Step 1: Check Solvability ($\mathbf{b} \in C(A)$ or $\mathbf{b} \perp N(A^T)$):**
   - Compute $\mathbf{y}^T\mathbf{b}$ for all basis vectors $\mathbf{y} \in N(A^T)$.
   - If any $\mathbf{y}^T\mathbf{b} \neq 0 \Rightarrow \mathbf{b} \notin C(A) \Rightarrow$ **0 Solutions (Inconsistent)**.
   - If all $\mathbf{y}^T\mathbf{b} = 0 \Rightarrow \mathbf{b} \in C(A) \Rightarrow$ **Solvable!** Proceed to Step 2.
2. **Step 2: Check Uniqueness (Nullspace Dimension $n - r$):**
   - If $r = n$ (Full column rank, $N(A) = \{\mathbf{0}\}$) $\Rightarrow$ **Exactly 1 Unique Solution** ($\mathbf{x} = \mathbf{x}_p$).
   - If $r < n$ (Free variables exist, $\dim(N(A)) > 0$) $\Rightarrow$ **Infinitely Many Solutions** ($\mathbf{x} = \mathbf{x}_p + \sum c_j\,\mathbf{s}_j$).